

SEYONG KIM

CURRENT POSITION

Ph.D. Student in Electrical and Electronic Engineering

Mar. 2023 – Present

- Yonsei University, Seoul, South Korea

CONTACT INFORMATION

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RESEARCH INTERESTS

Analysis and design of future wireless communication systems

- Advanced MIMO optimization
- Stochastic-geometry-based network modeling
- Low-resolution ADC/DAC architectures
- Physical-layer design for virtualized radio access networks (vRANs)

EDUCATION

Ph.D. in Electrical and Electronic Engineering

Mar. 2023 – Present

- Yonsei University, Seoul, South Korea
- Advisor: Prof. Jeonghun Park

M.S. in Electronic and Electrical Engineering

Mar. 2021 – Feb. 2023

- Kyungpook National University, Daegu, South Korea
- Thesis: “Downlink NOMA for Short-Packet Internet-of-Things Communications with Low-Resolution ADCs”
- Advisor: Prof. Jeonghun Park

B.S. in Electronic and Electrical Engineering

Mar. 2014 – Feb. 2021

- Kyungpook National University, Daegu, South Korea
- Degree period includes mandatory military service in South Korea (Mar. 2015 – Mar. 2017).

EXPERIENCE

Research Experience

Visiting Scholar

Mar. 2026 – Sep. 2026

- The University of Texas at Austin, Austin, TX, USA
- Host: Prof. Jeffrey G. Andrews

Work Experience

Project: Samsung Research Funding & Incubation Center of Samsung Electronics

2025 – 2026

- Funded by Samsung, Korea.
- Designed efficient space-time adaptive beamforming schemes tailored for high-mobility environments.

Project: Development of 5G-A vRAN Research Platform

2024 – 2026

- Funded by IITP, Korea.
- Analyzed the physical-layer implementation of NVIDIA pyAerial to evaluate its architectural alignment and

compliance with 3GPP standard specifications.

Project: 6G Cloud Research and Education Open Hub

2025 – Present

- Funded by IITP, Korea.
- Investigated schemes to enhance hotspot rate coverage using biasing and successive interference cancellation (SIC) in a multi-shell environment.
- Conducted this research in collaboration with Prof. Jeffrey G. Andrews.
- A manuscript is being prepared for submission to *IEEE Transactions on Wireless Communications*.

Project: Intelligent Secure Underwater Communications (InSeCT) Specialized Research Laboratory
2022 – 2025

- Conducted under the Defense Specialized Research Laboratory Program and funded by KRIT.
- Developed a link-level OFDM simulator and a Channel Quality Indicator (CQI) simulator tailored for underwater communication channels.

Teaching Assistant Experience

Teaching Assistant, Yonsei University, South Korea

- Design and Analysis of Advanced MIMO Systems Fall 2024
- Communication Systems Fall 2023
- Probability and Random Variables Spring 2023

Teaching Assistant, Kyungpook National University, South Korea

- Statistical Inference and Machine Learning Fall 2022
- Communication Theory Fall 2021, Fall 2022
- Information Theory Spring 2021
- Probability and Random Variables Spring 2021

AWARDS & HONORS

Outstanding Graduate Student Award (Top Prize)

2025

- Yonsei University, Seoul, South Korea

JOURNAL PUBLICATIONS

[3] **S. Kim** and J. Park, “Asymptotic Scaling Law Analysis of Multicast Satellite Communications with Massive MIMO”

- *IEEE Wireless Communications Letters*, 2025.

[2] **S. Kim**, J. Choi, W. Shin, N. Lee, and J. Park, “Multibeam Satellite Communications with Massive MIMO: Asymptotic Performance Analysis and Design Insights”

- *IEEE Transactions on Wireless Communications*, 2025.

[1] **S. Kim**, J. Choi, and J. Park, “Downlink NOMA for Short-Packet Internet-of-Things Communications with Low-Resolution ADCs”

- *IEEE Internet of Things Journal*, 2023.